The Information Order of Experiments: Intervention Lattices and Generator Identifiability

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Abstract

Which experiment should you run next? We give the question a finite, exactly computable form. For a hypothesis space $\mathcal{H}$ of candidate generators, write $O_1 \preceq O_2$ when O_1 is obtainable from O_2 by garbling. Refinement then contracts preimages, $P_{O_2}(g) \subseteq P_{O_1}(g)$, so per-generator ambiguity $i_O(g)$ and its population mean $H(G \mid O)$ both fall—a one-line consequence, but one that turns "more informative" into an order rather than a scale.

We instantiate the order exactly on all 1,500,625 four-state partially observed Markov chains, enumerating every subset of hidden-state resets. The reset sets form a Boolean lattice under inclusion; informativeness itself is only a preorder, and we reserve "lattice" for the former. Four findings. The passive experiment is a *deterministic* coarsening of the full interventional one, confirmed exactly, so its 1.874 bits of shortfall is precisely $I(G; O_I \mid O_P)$. Informativeness is genuinely partial, not total: a single reset leaves 4.504 bits where passive observation leaves 3.245, so one intervention is *worse* than watching, while two are better. Which two matters enormously—two resets inside one observational lump identify 0% of the class, two that cross it identify 58.55%, at identical cost—but crossing is not the rule. A sweep over all 15 partitions refutes it, and the exact criterion is conditional information: the value of adding reset j to a set A is $H(Z_j \mid Z_A) = H(Z_j) - I(Z_j; Z_A)$, intrinsic information less what is already known. And because the exact experiment outputs are deterministic functions of the generator, information gain is monotone submodular by Shannon's inequality, placing greedy selection on familiar footing.

The organising claim is that a finite generator-identification problem induces two complementary structures: an information order over experiments, governing contraction of generator preimages, and a belief-observability kernel, governing which hidden-state mixture directions remain invisible. The former yields an entropic polymatroid and a conditional-information selection rule; the latter yields a finite-horizon certificate for permanent observational blindness. We connect these views in one exactly enumerable setting without claiming novelty for the underlying Blackwell, submodular, or finite-rank machinery.

1 Introduction

A companion paper [12] argues that identifiability should be measured rather than assumed, by counting the preimage $P_O(g) = \{h \in \mathcal{H} : O(h) = O(g)\}$ of an observation map, and reports that identifiability and predictive uncertainty are distinct coordinates. It leaves a question open. Identifiability there is relative to a declared observation map and a declared class of admissible

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interventions, and changing that class moved 82% of one hypothesis space between regimes. If knowability is regime-relative, the regimes themselves need a structure.

They have one. Experiments are partially ordered by informativeness, and the order is old: Blackwell's comparison of experiments [1] ranks one experiment above another when the second can be obtained from the first by a stochastic garbling. What we add is a finite, exactly enumerable instance in which the whole lattice can be computed rather than argued about. Much of what earlier drafts of this paper reported as empirical has since become theorem or implementation check—submodularity, kernel contraction, the lumpable kernel dimension—and one conjecture was refuted outright. What survives as measurement is narrower and we state it plainly: that informativeness is a genuine partial order, with a single intervention less identifying than passive observation; that experimental redundancy tracks the dependency structure of the hypothesis class rather than its cardinality, so which resets are droppable changes with the constraint and not with $|\mathcal{H}|$; and that the belief-observability kernel can strictly contain the span of pairwise state equivalences, so a hidden-state *mixture* may be permanently invisible in a system that distinguishes every pair of states.

2 The refinement order

Let $\mathcal{H}$ be finite, G uniform on $\mathcal{H}$, and let O_1, O_2 be observation maps.

Definition 1. *$O_1 \preceq O_2$ iff there is a map F with $O_1 = F \circ O_2$.*

Proposition 2 (Containment)**.** *If $O_1 \preceq O_2$ then for every $g \in \mathcal{H}$, $P_{O_2}(g) \subseteq P_{O_1}(g)$, hence $i_{O_2}(g) \leq i_{O_1}(g)$ and $H(G \mid O_2) \leq H(G \mid O_1)$.*

Proof. If $O_2(h) = O_2(g)$ then $O_1(h) = F(O_2(h)) = F(O_2(g)) = O_1(g)$. Containment gives the inequality on $\log_2$ of cardinalities pointwise, and averaging preserves it. □

Proposition 3 (Value of refinement)**.** *If $O_1 \preceq O_2$ then $H(G \mid O_1) - H(G \mid O_2) = I(G; O_2 \mid O_1)$.*

Proposition 3 is what licenses reading a difference of conditional entropies as the information the finer experiment carries that the coarser one cannot reach. It requires the order; between incomparable experiments the difference is a comparison of two numbers and nothing more.

Proposition 4 (Submodularity)**.** *Let $Z_j = O_j(G)$ for j in a finite index set J, with each Z_j a deterministic function of G, and write $Z_A = (Z_j)_{j \in A}$. Then $F(A) = I(G; Z_A) = H(Z_A)$ is monotone and submodular in A. The same holds conditionally on any fixed observation P: $F_P(A) = I(G; Z_A \mid P) = H(Z_A \mid P)$ is monotone and submodular.*

Proof. $H(Z_A \mid G) = 0$ because each Z_j is a function of G, so $I(G; Z_A) = H(Z_A) - H(Z_A \mid G) = H(Z_A)$. The entropy of a collection of random variables is a monotone submodular set function, which is Shannon's inequality $H(Z_{A \cup \{j\}}) - H(Z_A) \geq H(Z_{B \cup \{j\}}) - H(Z_B)$ for $A \subseteq B$. Conditioning on a fixed P preserves both properties, and $I(G; Z_A \mid P) = H(Z_A \mid P)$ by the same argument. □

The pair (J, r) with $r(A) = H(Z_A)$ is therefore an *entropic polymatroid*, the dependence structure Fujishige identified for collections of random variables [2]. This is more specific than "some submodular objective": $r(\{j\}) = H(Z_j)$ is the standalone information of experiment j, $r(A \cup \{j\}) - r(A) = H(Z_j \mid Z_A)$ its marginal information, and $r(\{i\}) + r(\{j\}) - r(\{i, j\}) = I(Z_i; Z_j)$

the pairwise redundancy. Our contribution is not the greedy machinery, which is standard in observation selection [6], but that the finite-generator intervention problem induces an explicit entropic polymatroid whose rank values are exactly computable from generator equivalence classes, making those guarantees directly applicable.

Proposition 5 (One-way migration)**.** *Classify generators by thresholds on $i_O(g)$ and on a behavioural coordinate $p_k(g)$. If p_k is evaluated under an operating condition held fixed across $O_1 \preceq O_2$, then under refinement no generator can become less identifiable, and none can move in the behavioural coordinate.*

This is worth stating because it bounds what a migration statistic can mean. Reported migration under refinement is necessarily one-dimensional and one-directional; it is not evidence of two-dimensional mobility.

3 The benchmark system

We reuse the family of [12, §9] rather than introduce another: hidden state $X_t \in \{0,1,2,3\}$, transition rows drawn from $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ summing to one (35 rows), all $35^4 = 1{,}500{,}625$ generators enumerated completely, observation lumping $O(0) = O(1) = 0$, $O(2) = O(3) = 1$, horizon $T = 6$. The passive experiment starts from the uniform hidden state. For $A \subseteq \{0,1,2,3\}$, the experiment O_A resets the hidden state to each $j \in A$ in turn and returns the resulting observable laws.

4 Results

The passive experiment is a deterministic coarsening. Writing L_j for the law obtained by resetting to state j, we verified exactly, over every generator, that

$$O_P \;=\; \tfrac{1}{4} \textstyle\sum_j L_j,$$

so $O_P \preceq O_{\{0,1,2,3\}}$ with a deterministic F. Proposition 2 then applies, and we confirm its conclusion computationally as an implementation check: zero containment violations across every pair $A \subset B$, and zero against the passive mixture.

Reset set A	$H(G \mid O_A)$	classes	$\Pr[i_{O_A} = 0]$	$\max \lvert P \rvert$
$\varnothing$	20.517	1	0.00%	1,500,625
any singleton	4.504	400,584	0.00%	67,375
$\{0,1\}$ or $\{2,3\}$	3.356	482,902	0.00%	30,625
$\{0,2\},\{0,3\},\{1,2\},\{1,3\}$	2.618	967,107	58.55%	8,953
any triple	1.914	1,101,899	69.84%	7,263
$\{0,1,2,3\}$	1.371	1,235,809	82.01%	6,561
passive (mixture)	3.245	303,229	0.00%	7,537

Table 1: The intervention lattice, exact over 1,500,625 generators. Rows within a group are identical to the digits shown, reflecting the symmetry of the construction.

Informativeness is partial, not total. A single reset leaves 4.504 bits; passive observation leaves 3.245. One intervention is *less* identifying than watching a system you did not touch. There is no contradiction with Proposition 2: O_P is a coarsening of the full reset set, not of any singleton, so the two are incomparable in $\preceq$ and the proposition says nothing about them. Two resets (2.618) overtake passive; one does not. Three relations are in play and only two are comparabilities:

$$O_P \preceq O_{\{0,1,2,3\}}, \qquad O_{\{0\}} \preceq O_{\{0,1,2,3\}}, \qquad O_{\{0\}} \text{ and } O_P \text{ incomparable.}$$

Passive is a garbling of the full reset protocol, but neither passive nor a single reset is a garbling of the other, so their entropies may fall in either order and here they fall the surprising way. Any claim of the form "intervention beats observation" is false as stated; the correct statement is that informativeness is a partial order.

The interventions that pay are those crossing the observation's kernel. Let $x \sim_O y$ iff $O(x) = O(y)$, here $\{0, 1\}$ and $\{2, 3\}$. Two resets chosen inside one $\sim_O$ class leave 3.356 bits and identify *nothing*; two that cross leave 2.618 bits and uniquely identify 58.55% of the entire hypothesis class. The cost is identical. Greedy selection, run without knowledge of the lumping, chooses 0, then 2, then 3, then 1—crossing at the first opportunity.

We stated this as a conjecture—interventions crossing $\sim_O$ dominate interventions of equal cardinality confined within one class—and then tested it, with criteria fixed in advance, over *all* 15 set partitions of the four hidden states. It is false.

The sweep uses a coarser but still completely enumerated family so that every partition is affordable: transition rows from $\{0, \frac{1}{2}, 1\}$, giving $10^4 = 10{,}000$ generators, at $T = 5$; entropies are therefore not comparable in absolute terms with Table 1. Within each (partition, $|A|$) cell we asked whether mean $H(G \mid O_A)$ is non-increasing in the number of blocks A meets. It is in only 3 of 23 cells—13%, against a pre-committed threshold of 90%.

The failures have a clean shape. For the partition $\{0\} \mid \{1, 2, 3\}$ the effect *reverses*: two resets confined to the large block leave 4.852 bits, while two that cross leave 5.917—crossing is worse by 1.065 bits. For the balanced $\{0, 1\} \mid \{2, 3\}$ the original direction holds, 3.799 against 4.187. Restricting the test to partitions whose blocks are all the same size, it passes 3/3.

No replacement conjecture is required, because the exact rule is already available. Since each Z_j is a deterministic function of G, Proposition 4 gives $F(A) = H(Z_A)$, so the value of adding reset j to a set A is exactly its conditional novelty

$$\Delta(j \mid A) \;=\; H(Z_j \mid Z_A),$$

and for a pair, $H(Z_i, Z_j) = H(Z_i) + H(Z_j) - I(Z_i; Z_j)$. A pair of interventions is valuable when both carry information and little of it is shared. Block crossing was only ever a proxy for low redundancy, and decomposing the two cases (Experiment 25) shows they are governed by different terms.

On the balanced partition the individual entropies are *exactly equal*, so the entire 0.387-bit advantage of crossing is a redundancy effect: crossing pairs share less (4.854 against 5.241). On the unbalanced partition the redundancies are nearly equal —and the crossing pair is marginally the *more* redundant of the two—so essentially all of the 1.065-bit gap, 1.011 of it, comes from the individual term: a reset into a singleton observational block carries only 5.017 bits where a reset into the three-state block carries 6.028.

Partition	pair	crossing	$H(Z_i)+H(Z_j)$	$I(Z_i;Z_j)$	$H(G \mid O_A)$
$\{0\} \mid \{1,2,3\}$	$\{1,2\}$	no	12.056	3.621	4.852
	$\{0,1\}$	yes	11.045	3.675	5.917
$\{0,1\} \mid \{2,3\}$	$\{0,1\}$	no	14.342	5.241	4.187
	$\{0,2\}$	yes	14.342	4.854	3.799

Table 2: The crossing effect and its reversal, decomposed by $H(Z_i, Z_j) = H(Z_i) + H(Z_j) - I(Z_i; Z_j)$.

We had guessed the singleton reset was redundant. It is not; it is simply less informative on its own, which is a different mechanism and one the decomposition settles rather than argues. Crossing is a heuristic for low redundancy that works when symmetry equalises the individual terms and fails when it does not.

The selection rule that follows is exact rather than heuristic:

$$j^\star \;=\; \operatorname*{arg\,max}_{j \notin A} \; H(Z_j \mid Z_A),$$

which is the greedy step of Proposition 4 written in the variables that make it computable. What remains genuinely open is not the rule but its explanation: *which structural features of* $(\mathcal{H}, O)$ *make* $H(Z_j \mid Z_A)$ *large?* Observation block size re-enters there as a candidate explanatory variable rather than as the theory. A natural refinement to study is the interventional signature partition $x \sim_Z y \iff Z_x = Z_y$, which is not the observational partition $\sim_O$: states alike under $\sim_O$ but distinct under $\sim_Z$ are exactly the distinctions intervention can unlock, and states alike under both are hidden from this experimental family altogether.

We record the refutation rather than quietly narrowing the claim, because the conjecture as first written was the natural reading of Table 1 and would have been an easy thing to leave standing.

Information gain is submodular—by Proposition 4, not by observation. An earlier version of this paper reported 108 submodularity inequalities holding with zero violations and described the property as measured. That was a category error. In this setting the observable law of each reset is an exact deterministic function of the generator, so $F(A) = H(Z_A)$ and submodularity is Shannon's inequality; the computation could not have failed, and it checks our implementation rather than the mathematics. We record it as such: 108/108, consistent with the theorem.

What the theorem buys is operational. Maximising a monotone submodular function under a cardinality constraint admits the classical $(1 - 1/e)$ greedy guarantee [9], so under a budget $|A| \le k$ the rule

$$j^\star_t \;=\; \operatorname*{arg\,max}_{j \notin A_t} \; I\big(G; Z_j \mid P, Z_{A_t}\big)$$

—perform next the experiment expected to eliminate the most remaining ambiguity—is within $1 - 1/e$ of the best possible set of that size. That holds for any family of experiments whose exact outputs are deterministic in the unknown generator, which is the regime this framework assumes throughout. The greedy gain sequence here is 16.013, 1.886, 0.704, 0.543 bits; the fourth reset is nearly redundant.

The polymatroid has no non-trivial closure. A polymatroid carries a closure operator, $j \in \mathrm{cl}(A)$ iff $H(Z_j \mid Z_A) = 0$, and a basis is a minimal spanning set. We computed these exactly —$H(Z_J \mid Z_A) = 0$ holds iff A induces as many generator classes as J, an integer comparison rather than a tolerance—for all 14 non-trivial partitions of the four hidden states. In every one, the *only* spanning set is J itself. There is no experimental compression here: no reset is redundant given the other three, and a four-reset protocol cannot be replaced by a smaller one.

The margins by which this fails are informative. $H(Z_j \mid Z_{J\setminus\{j\}})$ ranges from 0.280 to 0.902 bits, and within every partition it is monotone in the size of j's observation block, with zero violations across all 14: a reset into a state the observation already isolates contributes least that the others cannot supply. Notably block size does *not* predict $H(Z_j)$ itself—Experiment 25 found those means non-monotone at 8.050, 8.157, 6.028 for block sizes 1, 2, 3—so the structural variable governs the conditional contribution and not the standalone one. This is the defensible residue of the intuition that failed in Experiment 24, and it is worth noting that the pairwise redundancies did not reveal it: $I(Z_0; Z_1)$ was marginally *lower* for the singleton in the reversal case, while the conditional-on-all-others measure separates cleanly.

A further contrast: the rank values $r(J)$ vary widely with the observation partition, from 9.892 to 13.288 bits, while the basis structure does not vary at all. What the observation map changes is how much the experiments tell you, not which of them are needed.

Compression exists, but it is a property of the hypothesis class. The absence of compression above is suspicious, because $\mathcal{H} = \mathrm{ROWS}^4$ is a product: the four transition rows are chosen independently, which is exactly the structure that should resist it. We therefore compared three classes on the same system and the same 14 partitions, with criteria fixed in advance. Alongside the product class we took a *tied* class constraining row 3 to equal row 0, giving 10^3 generators, and a *random* subset of the product class of the same cardinality. The random class is the control: it decides whether any compression is structural or merely an effect of having fewer generators to tell apart.

Hypothesis class	$\lvert\mathcal{H}\rvert$	partitions compressing	minimal bases
product, rows free	10,000	0/14	$\{0, 1, 2, 3\}$
tied, row 3 = row 0	1,000	14/14	$\{0, 1, 2\}$, $\{1, 2, 3\}$
random subset, same size	1,000	0/14	$\{0, 1, 2, 3\}$

Table 3: Compression is structural, not a cardinality effect: the tied class compresses in every partition while a random class of identical size compresses in none.

The tied class compresses in every partition, always to a basis of size three, and always by omitting one of the two tied states—never states 1 or 2, whose rows remain free. The equally sized random class compresses nowhere. So Experiment 26's result is a statement about the hypothesis class, not about interventions: *an intervention is redundant exactly when the generators carry a dependency that makes it so.*

It is tempting to read this as a clean separation—observation sets the rank magnitudes, hypothesis-class structure sets the basis—and an earlier version of this paper said so. Experiment 28 refutes it. Repeating the comparison across six classes and all 14 partitions, the basis is invariant under the observation map for the equality and adjacent ties, but *varies* for a permutation tie $R_3 = \pi(R_0)$ and for a two-row derived constraint $R_3 = f(R_0, R_1)$. So the observation map can change basis structure, for some dependency types and not others, and characterising which is open.

Which generator dependencies survive the observation map. A constraint among generator coordinates need not become a functional dependency among reset outputs, because Z_j is a function of the whole generator and not of row j alone. We made this precise with two closure operators, both tested by exact integer class counts: $\mathrm{cl}_G(A) = \{j : H(R_j \mid R_A) = 0\}$ among generator coordinates and $\mathrm{cl}_Z(A) = \{j : H(Z_j \mid Z_A) = 0\}$ among reset outputs.

Class	constraint	$\lvert\mathcal{H}\rvert$	minimal bases	basis invariant in O
FREE	none	10,000	$\{0, 1, 2, 3\}$	yes
EQUAL	$R_3 = R_0$	1,000	$\{0, 1, 2\}, \{1, 2, 3\}$	yes
PERM	$R_3 = \pi(R_0)$	1,000	$\{0, 1, 2\}, \{1, 2, 3\}$	**no**
DERIVED	$R_3 = f(R_0, R_1)$	1,000	$\{0, 1, 2\}, \{0, 2, 3\}, \{1, 2, 3\}$	**no**
ADJACENT	$R_2 = R_1$	1,000	$\{0, 1, 3\}, \{0, 2, 3\}$	yes
RANDOM	size-matched control	1,000	$\{0, 1, 2, 3\}$	yes

Table 4: Six hypothesis classes, five at identical cardinality. Which rows are tied determines which resets become droppable; a two-row constraint yields three bases rather than two.

Two pre-registered criteria failed, and both failures are informative. The inclusion $\mathrm{cl}_G(A) \subseteq \mathrm{cl}_Z(A)$ is *false*: 181 violations. The mechanism is transparent once seen—under $R_3 = R_0$ we have $3 \in \mathrm{cl}_G(\{0\})$, but Z_3 also depends on rows 1 and 2, so Z_0 alone cannot determine it. A coordinate dependency is not an output dependency.

The converse count was 0 in these six classes, and we briefly took $\mathrm{cl}_Z \subseteq \mathrm{cl}_G$ for an empirical regularity. It is not one. For deterministic experiment maps in general it fails at once: with R_0, R_1 independent and $Z_0 = Z_1 = R_0 \oplus R_1$, we have $1 \in \mathrm{cl}_Z(\{0\})$ while $1 \notin \mathrm{cl}_G(\{0\})$. Experiment 29 constructs the same effect inside the reset-law architecture.

The observation map can create dependencies: lumpability. Fix the balanced partition and constrain each row so its *block* transition sums depend only on the source state's block—states 0 and 1 both send mass a to block 0, states 2 and 3 both send b—while the split within each block stays free. Then R_0 and R_1 are not functions of one another, yet the observed process is Markov on blocks and resets 0 and 1 induce the same observable law. Characterising each class by $(\lvert\mathcal{P}\rvert, \lvert\mathcal{D}\rvert, \lvert\mathcal{C}\rvert)$—dependencies preserved, destroyed and created—under that partition:

Class	constraint	$\lvert\mathcal{H}\rvert$	$\lvert\mathcal{P}\rvert$	$\lvert\mathcal{D}\rvert$	$\lvert\mathcal{C}\rvert$
FREE	none	10,000	0	0	0
TIED	$R_3 = R_0$	1,000	8	0	0
LUMPABLE	block sums tied by block	1,156	0	0	**16**
TIED_LUMPABLE	both	340	8	0	**8**
RANDOM	size-matched control	1,156	0	0	0

Table 5: Redundancy has at least two independent origins. The lumpable class creates 16 dependencies with no generator dependency to inherit; a random class of the same cardinality creates none.

The created dependencies are exactly $Z_0 \leftrightarrow Z_1$ and $Z_2 \leftrightarrow Z_3$ and their extensions, and they appear under *one* of the 14 partitions—the balanced one the class was built for. Under any other observation the lumpability does not hold and nothing is created. The effect is the quotient matching the constraint, not the constraint alone.

A criterion for survival. The robust/fragile split is a taxonomy, not a criterion. Decomposing a reset law by its first symbol,

$$Z_s^{(T)} \;=\; \delta_{O(s)} \otimes \sum_x P_{sx}\, Z_x^{(T-1)},$$

makes the mechanism visible and suggests one. Under an identity tie $R_i = R_j$ the two sums agree term by term for every O and every T, so robustness is provable rather than observed; we ran it as an implementation check and found 0 counterexamples over 2002 generator–partition pairs. Under a permutation tie $R_j = \pi(R_i)$ the comparison is $\sum_y P_{iy} Z_{\pi(y)}^{(T-1)}$ against $\sum_y P_{iy} Z_y^{(T-1)}$, which agree when π moves each *reachable* destination to a behaviourally equivalent state, where $s \sim t$ iff $Z_s^{(T-1)} = Z_t^{(T-1)}$ for that generator. Tested in the sufficient direction over 2002 pairs:

	suffix laws equal	suffix laws differ
π behaviourally compatible	153	**0**
π incompatible	358	1491

Zero counterexamples: compatibility suffices. It is not necessary—358 cases agree without it—which is expected, since the sums can coincide numerically without the permutation respecting behaviour.

For the created case we pre-registered that dependencies appear exactly on partitions satisfying strong lumpability. That *failed*, on one partition: the discrete partition $0|1|2|3$ is strongly lumpable only vacuously, every block being a singleton, and creates nothing because there is nothing for the quotient to collapse. The refinement —lumpable *and* some block of size greater than one—matches dependency creation on all 14 partitions, and we record it as post-hoc.

One equation underneath all three. The three mechanisms are three cases of a single linear condition. Let b_x be the observable law of the next $T-1$ symbols from hidden state x, and $B_{O,T-1}(G)$ the matrix with those rows. Resetting to s and taking one transition leaves the hidden distribution P_s, so the suffix law is $S_s^{(T)} = P_s B$.

Proposition 6 (Belief-observability kernel)**.** *For a finite generator G, observation map O and horizon T,*

$$S_s^{(T)} = S_t^{(T)} \quad\Longleftrightarrow\quad (P_s - P_t)\, B_{O,T-1}(G) = 0 \quad\Longleftrightarrow\quad P_s - P_t \in \ker B_{O,T-1}(G).$$

Proof. $S_s = P_s B$ and $S_t = P_t B$, so $S_s = S_t$ iff $(P_s - P_t)B = 0$. □

Each earlier mechanism is a case. An *identity tie* has $P_s - P_t = 0$, which lies in every kernel—so its robustness across all O and T is immediate, and needs no appeal to behaviour. A *permutation tie* $P_t = P_s \Pi$ survives iff $P_s(I - \Pi)B = 0$, which is strictly weaker than the behavioural compatibility $\Pi B = B$ we tested in Experiment 30: compatibility forces each moved state's law to agree separately, whereas the kernel condition needs only their P_s-weighted mixture to cancel. That is what the 358 agreements without compatibility were—weighted cancellation, not coincidence. *Lumpability* is the case $P_i \neq P_j$ with $P_i - P_j \in \ker B$: the rows differ in a direction the observation is blind to. Verified over 11,004 reset pairs with zero failures, as an implementation check.

Proposition 7 (Horizon monotonicity)**.** $\ker B_{O,T+1} \subseteq \ker B_{O,T}$*, so* $\dim \ker B_{O,T}$ *is non-increasing in* T*.*

Proof. Deleting the final observation is a linear marginalisation M_T with $B_{T-1} = B_T M_T$. If $vB_T = 0$ then $vB_{T-1} = vB_T M_T = 0$. □

Corollary 8 (Full observation)**.** *If* O *is injective then* $\ker B_{O,T} = \{0\}$ *for every* $T \geq 1$*.*

Proof. Row x of B_T is supported on strings beginning with $O(x)$. For $x \neq y$ those first symbols differ, so the supports are disjoint and the rows are independent: $\operatorname{rank} B_T = n$. □

Proposition 9 (Lumpable quotient kernel)**.** *If* O *induces a* k*-block partition of* n *states and the chain is strongly lumpable with respect to it, then* $\operatorname{rank} B_{O,T} = k$ *and* $\dim \ker B_{O,T} = n - k$*, with kernel spanned by the within-block state differences* $e_x - e_y$*,* x, y *in a common block.*

Proof. Strong lumpability makes the observable law identical for states sharing a block, so rows within a block coincide, giving rank $\leq k$. Rows from different blocks are supported on strings with different first symbols, hence independent, giving equality. Rank–nullity supplies the dimension, and within-block differences are $n - k$ independent kernel vectors. □

Proposition 10 (Belief-space interpretation)**.** $K_T \subseteq \mathbf{1}^{\perp}$*: every belief-observability kernel direction has zero coordinate sum. Consequently, for an interior belief* $\mu \in \Delta^{n-1}$ *and* ϵ *small enough that* $\mu + \epsilon v$ *remains in the simplex,* μ *and* $\mu + \epsilon v$ *induce identical observable laws if and only if* $v \in K_T$*.*

Proof. Rows of B_T are probability laws over words, so $B_T \mathbf{1} = \mathbf{1}$. If $vB_T = 0$ then $0 = vB_T\mathbf{1} = v\cdot\mathbf{1}$. For the second claim, $(\mu + \epsilon v)B_T = \mu B_T$ iff $\epsilon v B_T = 0$ iff $vB_T = 0$. □

The kernel is therefore not merely algebraic. It is the nullspace of hidden-belief observability: the directions along which a belief can be moved without changing any observable future law. That notion is not ours. Van Handel [10] calls a hidden Markov model *observable* when no two initial measures of the signal induce the same law of the observation process, which is exactly the condition $K_\infty = \{0\}$ stated at the level of beliefs rather than of a kernel. Proposition 10 is that definition recovered constructively in finite dimensions, with an explicit basis and a horizon at which it is decidable. Writing

$$d_{\text{obs}} \;=\; (n-1) - \dim K_\infty$$

for the identifiable degrees of freedom of the hidden belief, the strongly lumpable 2+2 system has $d_{\text{obs}} = 3 - 2 = 1$: of three belief degrees of freedom only the mass assigned to the two aggregate blocks survives, which is one free parameter. Full row rank gives $d_{\text{obs}} = n - 1$, and all of belief space is observable.

This sharpens the mixture witness considerably. In that system—$P_3 = \frac{1}{2}(P_0 + P_2)$ under $\{0, 2, 3\} \mid \{1\}$—no two *pure* states induce the same observable law, yet

$$\nu = (\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}) \quad \text{and} \quad \mu = (\tfrac{1}{8}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{2})$$

are distinct beliefs in the simplex inducing *identical* observable laws, permanently, by Proposition 11. The statement is not that the kernel is richer than the state partition but the operational one: no two hidden states are observationally equivalent, and yet two different hidden beliefs are.

Blindness is structural or transient. $\dim\ker B$ measures how many directions of hidden transition behaviour the observation cannot see, and it separates the classes sharply. Two of its values are settled in advance. Under the discrete partition it is 0 for every generator of every class, as Corollary 8 requires; under the balanced partition the lumpable class has $\dim\ker B = 2$ for all 32 generators at horizons $T = 3$, 5 and 7 alike, with kernel spanned by exactly $e_0 - e_1$ and $e_2 - e_3$, which is Proposition 9 at $n = 4$, $k = 2$. The suspiciously exact 2.000 was exact because it had to be. We report both as implementation checks.

The contrast with an unconstrained class is the point. For FREE generators under the same partition the kernel dimensions are distributed $\{2:6,\ 1:17,\ 0:10\}$ at $T = 3$ and contract to $\{2:6,\ 1:14,\ 0:13\}$ by $T = 5$, then stop. Kernels contract with horizon by Proposition 7—measured non-increasing across $T = 2, \ldots, 6$ with zero exceptions, again a check rather than a finding—while the lumpable kernel does not contract at all, being pinned at $n - k$. There are two kinds of observational blindness: *transient*, removable by watching longer until it settles, and *structural*, which no horizon removes. That is the linear-algebraic counterpart of the preimage stabilisation at $T^\star$ reported in [12, §9]: preimages contract over candidate generators, kernels contract over hidden-state directions, and both reach a floor.

Proposition 11 (Finite determination)**.** *Let the hidden state set have n elements and let O have k non-empty blocks. If $K_{T+1} = K_T$ for some T then $K_s = K_T$ for all $s \geq T$; and writing $T_{\text{obs}} = \min\{T : K_T = K_\infty\}$,*

$$T_{\text{obs}} \ \leq\ n - k + 1, \qquad \textit{so} \qquad K_\infty = K_{n-k+1}.$$

Proof. Let E_y be the diagonal selector for symbol y and, for a word $w = y_0 \cdots y_{T-1}$, put $q_w = E_{y_0} P E_{y_1} P \cdots P E_{y_{T-1}} \mathbf{1}$, whose x-th entry is $\Pr(Y_{0:T-1} = w \mid X_0 = x)$. The columns of B_T are the q_w, so $K_T = V_T^\perp$ with $V_T = \operatorname{span}\{q_w : |w| = T\}$. Since $q_{yw} = E_y P q_w$ we have $V_{T+1} = \operatorname{span}_y E_y P\, V_T$. Hence if $V_T = V_{T-1}$ then $V_{T+1} = \operatorname{span}_y E_y P V_{T-1} = V_T$, and by induction the chain is constant from T onward; dualising gives the first claim. For the bound, $V_1 = \operatorname{span}\{E_y \mathbf{1}\}$ has dimension exactly k because those vectors have disjoint supports, and $V_T \subseteq V_{T+1}$ with equality triggering permanent stabilisation, so each step before stabilisation strictly increases dimension inside an n-dimensional space. At most $n - k$ such increases are available. $\square$

Two consequences are worth stating. For injective O we have $k = n$ and the bound reads $T_{\text{obs}} \leq 1$, recovering Corollary 8. For the four-state balanced lumping used throughout, $n = 4$ and $k = 2$ give $K_\infty = K_3$ exactly—so the stabilisation observed by $T \leq 3$ in every case we sampled was not a property of the sample. Permanent semantic blindness in this setting is finitely decidable, and the horizon needed scales as hidden dimension minus immediately visible dimension.

The bound is attained. Checking 910 generator–partition cases through horizon 8, no T_{obs} exceeds $n-k+1$ and no plateau is later broken; and for each k the maximum is reached—$T_{\text{obs}} = 3$ occurs 94 times at $k = 2$, $T_{\text{obs}} = 2$ occurs 265 times at $k = 3$, and every case at $k = 4$ has $T_{\text{obs}} = 1$. So the bound is tight rather than merely valid, and no smaller horizon suffices in general.

The proof does not in fact use determinism, and saying so sharpens the result. Nothing in the plateau argument needs E_y to be a 0/1 selector; it needs only $\sum_y E_y = I$ and $P\mathbf{1} = \mathbf{1}$, both of which hold when $E_y = \operatorname{diag}\big(\Pr(Y = y \mid X = x)\big)$ for a stochastic emission matrix $C_{xy} = \Pr(Y = y \mid X = x)$. Only the starting dimension changes: $E_y\mathbf{1}$ is the y-th column of C, so $\dim V_1 = \operatorname{rank} C$.

Corollary 12 (Stochastic emissions)**.** *For an n-state time-homogeneous finite HMM with emission matrix C, the kernels satisfy $K_{T+1} \subseteq K_T$, any one-step plateau is permanent, and*

$$K_\infty \;=\; K_{n-\operatorname{rank}(C)+1}.$$

Checking the generalisation over 15 transition matrices and emission matrices of each rank, no T_{obs} exceeds the bound, and it is attained for $\operatorname{rank} C = 2, 3, 4$. It is *not* attained at rank 1, and that gap is structural rather than a sampling artefact: a row-stochastic C of rank 1 has all rows equal, since they are proportional and each sums to one, so emission is independent of state, $V_1 = \operatorname{span}\{\mathbf{1}\}$, and $V_2 = \operatorname{span}_y E_y P\mathbf{1} = \operatorname{span}_y E_y\mathbf{1} = V_1$. Thus $T_{\text{obs}} = 1$ whenever $\operatorname{rank} C = 1$, and the bound n is unattainable for $n > 1$. The bound is therefore tight for $\operatorname{rank} C \geq 2$ and loose exactly at the degenerate rank where the observation is uninformative.

Proposition 11 is the case where C is a 0/1 matrix with one entry per row, whose columns are the block indicators, so $\operatorname{rank} C = k$. The controlling quantity is therefore not the size of the observation alphabet but the rank of the one-step emission information: an alphabet with many symbols whose state signatures span a low-dimensional subspace behaves like a coarse partition, and an emission matrix of full row rank gives $T_{\text{obs}} = 1$ without any deterministic state label. We claim Corollary 12 as a transparent rank-sensitive finite-horizon certificate inside this framework, not as a new finite-equivalence principle. The underlying algebra is the reachable-subspace argument familiar from probabilistic and weighted automata, where equivalence is decidable in polynomial time [11], and from observable operator models [5] and predictive state representations [7], which organise dynamical systems by exactly this finite linear dimension. Proposition 10 is the observability statement those frameworks make central, recovered here in the notation of the preimage framework.

We state these results for the model formalised here: finite hidden-state set, finite observation alphabet, time-homogeneous transitions, exact finite-dimensional laws. We do not extend them to controlled or state-action-dependent observation, nonstationary dynamics, continuous hidden state, or empirically estimated laws, where the exact finite-dimensional linear representation is unavailable.

The kernel is strictly richer than a state partition. Classical behavioural equivalence of states, $x \equiv y$ iff they induce identical observable laws, embeds in the kernel as the difference vectors $e_x - e_y$. Write D for their span; $D \subseteq \ker B$ always. The question is whether the inclusion is strict, and it is: over 462 generator–partition pairs from the unconstrained family, 4 have $\dim\ker B > \dim D$. More pointedly, in 3 cases two generators share both the observation partition and the state-equivalence classes—all singletons, no two states equivalent—while differing in $\dim\ker B$, which shows $\ker B$ is not a function of the partition.

One witness can be checked by hand. Under the partition $\{0, 2, 3\} \mid \{1\}$, take

$$P_0 = (0, 1, 0, 0), \quad P_1 = (0, 1, 0, 0), \quad P_2 = (1, 0, 0, 0), \quad P_3 = (\tfrac{1}{2}, \tfrac{1}{2}, 0, 0).$$

No two states induce the same observable law, so $D = \{0\}$. But $P_3 = \frac{1}{2}(P_0 + P_2)$, and states 0, 2, 3 share a first observation symbol, so $b_3 = \frac{1}{2}(b_0 + b_2)$ and $v = (-\frac{1}{2}, 0, -\frac{1}{2}, 1)$ lies in $\ker B$ at every horizon. The observation is blind to a *mixture* of hidden states while distinguishing every pair of them.

This is why the belief-observability kernel is the right object rather than a refinement of the state partition. A partition can only record which states are indistinguishable; the kernel records which linear combinations are, and the 358 weighted cancellations of Experiment 30 are of exactly this kind. The kernel dimension stabilised at $T \leq 3$ throughout—142, 261 and 59 pairs at $T = 1, 2, 3$—which Proposition 11 says had to happen, since $n - k + 1 = 3$ here. The witness above therefore certifies more than it appears to: exhibiting $v \in K_3$ for a two-block observation of four states establishes $v \in K_\infty$, so that mixture is invisible at *every* horizon, not merely at the ones we computed.

Proposition 6 also gives the robust/fragile distinction a definition rather than a label: a dependency is *robust* when its difference vector lies in $\ker B_{O,T}$ for every admissible O, which identity ties satisfy because that vector is zero, and *fragile* when membership varies with O.

Robust and fragile dependencies. The two failures of Experiment 28 turn out to be one phenomenon. Breaking its 181 inclusion violations down by class gives 100 for PERM and 81 for DERIVED, and zero for every other class—and those are precisely the two classes whose bases varied with the observation map. A dependency that survives every observation map cannot move the basis; a dependency some observation maps destroy must. So the constraint types split into *robust* (the identity ties EQUAL and ADJACENT: nothing destroyed, basis invariant) and *fragile* (PERM and DERIVED: dependencies destroyed, basis varies), with lumpability supplying a third category that owes nothing to the generator at all.

The value of full intervention. $H(G \mid O_P) - H(G \mid O_I) = 3.245 - 1.371 = 1.874$ bits, which by Proposition 3 is exactly $I(G; O_I \mid O_P)$: the generator information the complete reset protocol carries that no amount of passive observation can reach.

5 Related work

The order is Blackwell's [1]; our instance is its degenerate deterministic case. Choosing experiments to maximise expected information is Lindley's [8], and choosing interventions to secure identifiability is well developed in causal discovery, where optimal strategies are known [3]. The greedy guarantee for submodular objectives is [9]. We claim no new theory of Blackwell comparison, submodular information selection, probabilistic-automaton equivalence, or predictive-state observability; each is established and cited above. The contribution we do claim is integration: an exactly enumerable generator-identification framework in which these become different views of one experimental problem—observation induces generator preimages, refinement orders those preimages, intervention outputs induce an entropic polymatroid, and the belief-observability kernel characterises the hidden-belief directions no observation horizon can distinguish. What we contribute technically is an exactly enumerated lattice: every experiment in a finite family, every containment relation checked rather than assumed. Of the facts we first took to be empirical, submodularity is a theorem (Proposition 4) and the crossing rule is refuted; what survives as measurement is the incomparability of partial interventions with passive observation, the origins of redundancy, and the kernel dimensions outside the cases Propositions 7–9 settle.

6 Limitations

The lattice is computed on one family with one lumping, and the striking symmetry of Table 1 is a property of that construction. One comparison is outstanding and should be settled before this is circulated widely. Ito, Amari and Kobayashi [4] solve identifiability for hidden Markov processes by linear algebra and introduce a Δ-subspace characterising the effective minimum degree of the process. We attempted this comparison and could not complete it: the construction is behind a paywall in both the IEEE and the Electronics and Communications in Japan printings, and the open-access sources we reached describe the result only as "identifiability is completely solved by linear algebra, where a block structure of the transition matrix plays a fundamental role" without giving the subspace definition. The comparison therefore requires the primary text and remains undone. On the evidence available it is entirely possible that the belief-observability kernel is a dual, annihilator, or otherwise equivalent presentation of an object already implicit there, and a reader with access should treat establishing that as the first task. Nothing in this paper should be read as claiming the kernel notion is new; the wording throughout is that it is recovered in the notation of the preimage framework.

We do not know what distinguishes robust from fragile dependencies beyond the four examples in Table 4; identity ties are robust and permutation or two-row constraints are fragile, which is a pattern with $n = 4$. The lumpable construction of Table 5 is one aggregation matched to one partition, and we have not characterised which quotients create dependencies for which classes. The crossing conjecture is refuted in general and survives only for equal-block partitions; the exact selection rule $\Delta(j \mid A) = H(Z_j \mid Z_A)$ replaces it, but we have no structural characterisation of when that quantity is large. The kernel machinery does not answer it: the two sides of this paper leave two distinct open problems, and it would be a mistake to read the second as having settled the first. On the generator side, we do not know what structural features of $(\mathcal{H}, O)$ make the conditional novelty $H(Z_j \mid Z_A)$ large, which is the experiment-value question. On the belief side, we do not know how K_∞ relates to the minimal-realisation subspaces of the hidden Markov literature, which is an observability question. They are connected—both are driven by what the observation erases—but they are not the same object and neither reduces to the other. Submodularity is a theorem here (Proposition 4) and holds whenever exact experiment outputs are deterministic functions of the generator; it would need revisiting for noisy or sampled observations, where $H(Z_A \mid G) > 0$ and the identity $F(A) = H(Z_A)$ fails. Identification is at horizon $T = 6$, which is where the partition of this family stabilises [12, §9], but stability over a finite window is evidence rather than proof. Proposition 2 is a theorem, so the computation confirming it tests our implementation, not the mathematics.

7 Discussion

Resolving a mechanism is partition refinement. Beginning with $[g] = \mathcal{H}$, each experiment induces an equivalence relation and the cell containing the truth contracts,

$$\mathcal{H} \supseteq [g]_{\Pi_1} \supseteq [g]_{\Pi_2} \supseteq \cdots ,$$

with perfect resolution at $[g]_\Pi = \{g\}$ and irreducible ambiguity when the admissible order bottoms out while $|[g]_\Pi| > 1$. Three structures sit on top of one another: generator space is partitioned by an observation; the information order says which partitions refine which; and submodularity controls how efficiently refinements can be bought under a budget. What the measurements add is that the rate of contraction is controlled by the order on experiments, that the order is partial rather than total—so more control is not uniformly more information—and that the value of an additional experiment is the conditional information it contributes beyond those already performed.

Reproducibility. `mgs_lattice.py` produces the intervention-order figures and `mgs_partitions.py` the partition sweep of §4; `audit.py` re-derives every number here against them, and reports its status in `AUDIT.txt`. All are in the repository accompanying [12].

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